

COMSM0067: Advanced Topics in Programming Languages
Problem Sheet 8

Hoare Logic I — triples and the rules

1. Draw derivations that justify the following triples, naming every rule used.

(i) $\{x = 3\} \ y := x + 1 \ \{y = 4\}$

(ii) $\{\text{True}\} \ x := 1; \ y := x + 1 \ \{y = 2\}$

(iii) $\{x = A\} \ \text{if } x \leq 0 \text{ then } y := -x \text{ else } y := x \ \{y = |A|\}$

2. Which of the following triples are *valid*? For each, either give a derivation or exhibit a state that refutes it.

(i) $\{x = 6\} \ x := 5 \ \{x = 5\}$

(ii) $\{\text{True}\} \ x := 5 \ \{x = 6\}$

(iii) $\{x > 0\} \ \text{while } x > 0 \text{ do } x := x + 1 \ \{x = -17\}$

(iv) $\{\text{False}\} \ c \ \{Q\}$ for arbitrary c and Q

3. (*) Prove that the following program meets its specification. There is no loop, so the assignment axiom and SEQ do all the work: start from the postcondition and push it backwards through the program, writing down the assertion you obtain at each step.

$\{x = A \wedge y = B\} \ \text{if } x > y \text{ then } (t := x; \ x := y; \ y := t) \text{ else skip } \{x \leq y \wedge x + y = A + B\}$

Say which rule you use to bring the two branches of the if to a common postcondition.